

MANAV RUNTHALA

Software Engineer · AI/ML & Full-Stack Systems Builder · B.Tech CSE AIML @ JECRC × IBM (2028 Batch)

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SUMMARY

B.Tech CSE (AIML) student at JECRC University (IBM Collaboration, 2028 batch) with hands-on experience designing and implementing production-grade, end-to-end software systems. Built 5 full-stack applications spanning distributed backends, scalable data pipelines, real-time web crawlers, and AI-integrated recommendation engines using Python, JavaScript, and Rust. Strong foundation in data structures, algorithms, software design, debugging, and system reliability. Seeking Google Software Engineering Internship (Summer 2027) to build scalable solutions for large-scale systems.

TECHNICAL SKILLS

Languages: Python, JavaScript, Rust (systems engine), SQL, Bash/Shell — experienced in general purpose programming for full-stack and backend development

Data Structures & Algorithms: Arrays, Hash Maps, Queues, Trees, Graphs; implemented in Python across ML pipelines and crawler deduplication systems

AI / ML & Computer Science: Machine Learning, Deep Learning (TensorFlow), Dimensionality Reduction, Spark MLlib, Text Analytics, NLP, IBM Watson Studio, Data Mining, Computer Vision fundamentals

Distributed Systems & Backend: Multi-worker architectures, cron scheduling, async task queues, REST APIs, Node.js/Express, PM2 process management, scalability patterns

Data & Visualization: Microsoft Power BI, Microsoft Fabric, Business Intelligence, KPI dashboards, Statistics, data visualization and analytics

DevOps, Debugging & Reliability: Docker, Docker Compose, GitHub Actions CI/CD, Vercel, performance monitoring, debugging, system reliability, Dockerized networking

Cloud & Infrastructure: AWS Cloud Practitioner Essentials (In Progress), Microsoft Azure AZ-900 (In Progress), cloud architecture concepts

Soft Skills: Collaboration, communication, teamwork, leadership, analytical thinking, innovation, creativity

PROJECTS

GlitchOver Creator Finder — Production-Deployed AI Outreach Engine · Python · JavaScript · Docker · Vercel · GitHub Actions CI/CD · Twitch API · ML [[GitHub](#)]

- ▶ Designed and implemented an end-to-end AI-powered platform that discovers, scores, and targets content creators across Twitch and social platforms using API integrations and ML-based reply prediction
- ▶ Built full-stack system: Python backend (69%), JavaScript frontend (22%), Dockerized with multi-service networking, automated CI/CD pipelines via GitHub Actions for reliable continuous deployment
- ▶ Shipped to production on Vercel — 325+ deployments, 366 commits, 5 branches, 2 tagged releases; implemented scalability patterns including caching, deduplication, and creator profile scoring algorithms
- ▶ Engineered daily targeting engine with CSV-based bulk processing, debugging-first architecture, and data structures (hash maps, deduplication indexes) to handle large creator datasets efficiently

Student Support Chatbot Backend — Structured Node.js Backend System · Node.js · Express · REST API · Cron · Automated Testing · GitHub Actions [[GitHub](#)]

- ▶ Architected a modular, production-ready Node.js backend for a student support chatbot — 215 commits across models, routes, utils, cron, tests, and workflow layers with full separation of concerns
- ▶ Implemented FAQ intelligence engine using keyword-based data structures (knowledgeDiary.json, keywordFaqs.json), chat logging, admin routes, and automated test suites for software reliability
- ▶ Designed for scalability and reliability with scheduled cron jobs, GitHub Actions CI/CD workflows, and admin-level debugging and monitoring routes

Astro Trading Engine — AI-Powered Financial Signal System · Python · Rust · ML Models · Financial APIs · PM2 · Full-Stack · Workers

- ▶ Built an AI-integrated financial engine combining planetary position data with market signals to generate daily trading sentiment scores — unique intersection of data mining and financial analytics
- ▶ Implemented high-performance Rust systems engine for core computation alongside iterative Python ML workers (V2+V3 models), demonstrating multi-language architecture and performance optimization
- ▶ Full-stack distributed architecture: frontend app, Python backend, Rust core engine, worker processes, PM2 process manager for persistent backend reliability, config/scripts/tests layers

Credit Card Intelligence Engine — AI Recommendation Platform · Python · JavaScript · Docker · Docker Compose · REST API · Full-Stack [[GitHub](#)]

- ▶ Developed an end-to-end intelligent recommendation system that analyzes user financial profiles and spending patterns to match optimal credit card products using algorithmic scoring
- ▶ Full-stack architecture: Python (64%) + JavaScript (34%), Dockerized with docker-compose for multi-service networking, structured across backend/frontend/data/gateway with cloud-ready Procfile deployment

- ▶ Delivered 3-phase stabilization lifecycle (phases 1-3) with premium dashboard upgrade and optimization engine — 33 commits demonstrating iterative software engineering and debugging discipline

TradeLead AI — Global Import/Export Discovery Engine · Python · Distributed Crawlers · Worker Processes · Data Pipelines · Storage · REST API

- ▶ Built a keyword-driven global importer discovery system with distributed crawlers that extract verified business websites and emails for export lead generation — scalable data pipeline architecture
- ▶ Designed modular distributed system: crawler, discovery, extraction, workers, storage, and export modules — independent components enabling scalability, reliability, and easy debugging per service
- ▶ Implemented multi-stage data processing with deduplication algorithms, filtering logic, and structured CSV/JSON export — designed for performance with large-scale trade data across global markets

EXPERIENCE

B.Tech CSE AIML Project Student — IBM Collaboration Program — JECRC University, Jaipur Jul 2024 – Present

- ▶ Completed 13+ IBM-verified certifications in Data Science, ML, Deep Learning (TensorFlow), NLP, Spark MLlib, IBM Watson Studio, AI Services, and Cloud Computing
- ▶ Applied computer science fundamentals to real-world challenges: built production-deployed systems integrating distributed backends, ML models, REST APIs, and full-stack architecture
- ▶ Collaborated on team-based AI/ML projects; developed expertise in Python programming, data structures, debugging, data visualization, and end-to-end software engineering

EDUCATION

B.Tech — Computer Science & Engineering (AI & Machine Learning) — JECRC University × IBM, Jaipur Jul 2024 – Jun 2028

- ▶ IBM-partnered AIML specialization; 2028 batch — eligible for Google Summer 2027 Internship
- ▶ Core coursework: algorithms, data structures, machine learning, deep learning, data analytics, Python, cloud computing, statistics, software design

Class 12 — Science | 70% — ST Jesus and Mary School, Bikaner May 2024

Class 10 | 86% — Delhi Public School, Bikaner May 2021

CERTIFICATIONS

IBM / JECRC SkillsBuild — 13 Verified Certificates: Deep Learning with TensorFlow · Machine Learning – Dimensionality Reduction · Spark MLlib · Data Science 101 · Data Science Methodology · Data Visualization · Introduction to Python · Text Analytics 101 · Rapid Development for AI (AI Services) · IBM Watson Studio · Cognitive Decision System for Managers · ML with Sound · Statistics 101

Cloud & Microsoft (In Progress): AWS Cloud Practitioner Essentials · Microsoft Azure Fundamentals (AZ-900) · Analyze Data with Microsoft Fabric · Get Started with Power BI · Microsoft Data Analytics (5 Learn Achievements)

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